



TASHENT UNIVERSITY OF TECHNOLOGY

Faculty of Energy Engineering

1. Subject/Module Information

Field	Content
Subject/Module Name	Introduction to Engineering
Subject/Module Code	CC-INEN010104
Subject/Module ECTS	4
Academic year	2026 - 2027
Study Semester	Fall (semester 1)
Subject/module type	Mandatory
Program Affiliation	Bachelor of Science for all tracks
Module Coordinator(s)	Dr. Mohammad Reza Bahrani Dr. Khamidov Ibrokhimjon

2. Forms of Learning and Teaching

Workload breakdown (30 hrs/ECTS convention):

Component	Hours
Total Classroom Hours	60
Lecture	30
Project Work (team-based, in-session)	30
Independent Study	60
Total Workload	120

3. Module Description

Introduction to Engineering introduces students to the culture and identity of the engineering profession, emphasizing the role of engineering in historical, social, and economic development. The course provides a structured understanding of the engineering method — spanning problem definition, modelling, quantitative reasoning, and continuous improvement — and promotes ethical awareness and social responsibility in technological decision-making. The course adopts a Project-Based Learning (PBL) approach: rather than covering a specific technical domain, it develops core conceptual building blocks common to all fields of engineering — systems thinking, problem framing, quantitative reasoning, decision-making under trade-offs, and simulation — and applies them continuously within a single team-based project carried through the full semester.

4. Entry Requirements

Field	Content
Prerequisites	None; the course is accessible to all first-year students regardless of prior background.
Co-requisites	No
Additional Remarks	Students are expected to engage with basic quantitative reasoning, participate actively in team-based work, and communicate ideas clearly in spoken and written English.

5. Intended Learning Outcomes (ILOs)

Each ILO carries a Bloom level and is mapped to ASIIN's four competence dimensions — Subject-specific (Fachkompetenz), Methodological (Methodenkompetenz), Systemic (Systemkompetenz), Social (Sozialkompetenz).

No	Bloom Level	ILO	ASIIN Dimension
1	Understand	Explain that engineering models are purposeful simplifications of complex systems, and describe how assumptions and approximations shape a model's usefulness and limitations.	Subject-specific
2	Apply	Apply basic quantitative reasoning — demand, capacity, rates, flows, order-of-magnitude estimation — to characterize the behavior of a real-world system using limited or incomplete data.	Subject-specific
3	Analyze	Analyze system dynamics such as delays, queues, throughput, and bottlenecks, and use simple simulations to explore how a system behaves under uncertainty and varying assumptions.	Methodological
4	Evaluate	Evaluate trade-offs among competing objectives — efficiency, cost, fairness, accessibility — and justify a design decision that accounts for technical, ethical, and stakeholder considerations.	Systemic
5	Apply	Collaborate effectively within a team on a semester-long project, and communicate models, assumptions, and decisions clearly in written and oral form to peers and instructors.	Social

6. Course Content (Weekly Breakdown)

Week	Topic	Format	Hours
1	Engineering as a profession — culture, identity, and the historical, social, and economic role of engineering. How engineering knowledge differs from (and connects to) scientific inquiry and design.	Lecture + Project	4
2	Systems thinking I — systems as interacting elements, processes, and stakeholders; system boundaries and operating context.	Lecture + Project	4
3	Systems thinking II — process flow diagrams; how different perspectives lead to different interpretations of the same system.	Lecture + Project	4
4	Problem framing & modelling I — transforming informal observations into a well-defined engineering problem; identifying relevant variables.	Lecture + Project	4
5	Problem framing & modelling II — engineering models as simplified representations; assumptions and their consequences.	Lecture + Project	4
6	Quantitative reasoning I — demand and capacity, rates and flows, order-of-magnitude estimation.	Lecture + Project	4
7	Quantitative reasoning II — reasoning with limited or incomplete data; comparing interpretations of the same problem.	Lecture + Project	4
8	Midterm	Assessment	4
9	Dynamic system behavior — delays, queues, throughput, bottlenecks; introduction to spreadsheet-based modelling tools.	Lecture + Project	4
10	Engineering decision-making & trade-offs — balancing efficiency, cost, fairness, and accessibility; there is rarely one “optimal” solution.	Lecture + Project	4
11	Uncertainty & variability — “what-if” analysis; assessing robustness and reliability of a system and its model.	Lecture + Project	4
12	Simulation as a way of thinking — extending qualitative models into simple quantitative simulations to compare scenarios.	Lecture + Project	4
13	Ethics, stakeholders & responsible engineering — fairness, safety, accessibility, and sustainability as integral to design and decision-making.	Lecture + Project	4
14	Communicating engineering — technical writing and presentation skills; project synthesis workshop.	Project	4
15	Overview of the course	Lecture	4

7. Independent Study Assignments*

Week	Assignments	Hours
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1	Individual reflection: identify three everyday systems and describe their boundaries and stakeholders, in preparation for team system selection.	4
2	Team task: finalize choice of system; sketch initial system boundary and stakeholder map (Milestone 1 draft).	4
3	Team deliverable: complete and submit process flow diagram and stakeholder map (Milestone 1).	4
4	Team task: draft problem statement and a working definition of “system performance” for the chosen system.	4
5	Team deliverable: submit problem framing document and explicit list of modelling assumptions (Milestone 2).	4
6	Individual problem set: order-of-magnitude estimation and rate/flow exercises applied to the team's system.	4
7	Team deliverable: first-pass quantitative model of the system; prepare a short in-class pitch (Milestone 3).	4
8	Individual self-assessment: reflect on instructor/peer feedback from the Milestone Review and revise the team work plan for the second half of the semester.	4
9	Team task: build a spreadsheet model capturing delays, queues, or bottlenecks in the system.	4
10	Team deliverable: trade-off analysis comparing at least two alternative designs (Milestone 4).	4
11	Team task: stress-test the model against varying assumptions; document sensitivity of results.	4
12	Team deliverable: draft scenario simulation and comparison of outcomes (Milestone 5).	4
13	Team task: integrate ethical and stakeholder analysis into the project narrative.	4
14	Team task: draft final report and rehearse final presentation; incorporate peer critique.	4
15	Individual task: complete peer review of teammates' engagement and contribution.	4

***Note:** Independent study assignments are structured around the continuous team project carried through the full semester. All group tasks (at least one-third of assignments, per university convention) support the team's single system/project; individual tasks support reflection, preparation, and peer assessment.

8. Teaching and Learning Methods

- Lectures / practical classes introducing each conceptual building block
- Team-based project studio sessions (in-class project work)
- Structured milestone deliverables with instructor and peer feedback
- Team presentations and peer review
- Optional guest speaker session / industry site visit

9. Assessment Methods and Completion

Component	Examination Type	Duration/Length	Weight (%)	ILOs
Weekly mini projects	Report + presentation	30 minutes	30	1–4
Midterm Exam	Paper based examination	1-2 hours	30	1–4
Final Exam	Paper based examination	1-2 hours	40	6

Grading and Completion

90-100 points = A - Excellent
75-89 Points = B - Good
60-74 Points = C - Satisfactory
0 - 59 Points = D - Poor

10. Indicative Literature

- P. Kosky, R. Balmer, W. Keat, G. Wise, Exploring Engineering: An Introduction to Engineering and Design, Academic Press, 2020.
- Supplementary case studies and readings provided by the instructor, as assigned throughout the semester.

11. Statement on Plagiarism and Academic Dishonesty:

Tashkent University of Technology observes a zero tolerance policy regarding academic dishonesty. Per university policy as described in the academic catalog, all cases of academic dishonesty will be reported. The following policy will be used for this course:

1. First instances of academic dishonesty will result in a 0 grade on the assignment, but a second attempt may be allowed at the instructor's discretion.
2. Second instances of academic dishonesty will result in a 0 grade on the assignment with no opportunity to reattempt.
3. Third instances of academic dishonesty will result in automatic failure of the course.

12. Statement on the use of AI-tools

Tashkent Tech recognizes artificial intelligence (AI) as a powerful tool for learning, innovation, and skill development. Students are encouraged to use generative AI tools (such as ChatGPT, Claude, DeepSeek or similar platforms) to clarify concepts, debug code, and improve their learning through dialogue (with the AI tools). When used responsibly, AI can serve as a valuable collaborative partner. However, any content generated by AI must be treated as a source, meaning that students must explicitly cite the tool used and clearly document how it assisted you in their workflow, in accordance with each course's specific guidelines.

Academic integrity remains the fundamental principle governing AI use. Students may not submit AI-generated text, code, calculations, or other outputs as their own work unless explicitly authorized by the instructor. All graded assignments, essays, and exams must reflect the student's own understanding and be explainable in detail upon request. The unauthorized or unacknowledged use of AI to complete assignments will result in academic penalties, as stated in the *Statement on Plagiarism and Academic Dishonesty* guidelines.

Attendance Requirements: **Per University policy, students must attend at least 75% of class meetings to be allowed to take the final exam. If you miss 25% or more of class meetings, you will not be allowed to take the final exam.**